

Deep Operator Learning for Viscoelastic Constitutive Modeling (Neural UMAT)

Scientific Machine Learning-Project Work

Midhun Mohanan Maralakuzhiyil

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Supervisor: Mr. Kristian Boroz

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1 Viscoelastic Constitutive Modeling (DeepONet)

1.1 Background and Motivation

In multiscale mechanics, predicting the macroscopic behavior of heterogeneous materials (e.g., composites) requires solving the micro-scale boundary value problem at every integration point of the macroscopic mesh. This concurrent multiscale approach (often termed FE^2) is computationally prohibitive. For a typical mesh with 10^6 elements and 10^3 time steps, a direct numerical simulation would require 10^9 micro-scale solves.

To overcome this, we propose replacing the expensive micro-scale solver with a neural surrogate. However, standard neural networks (MLPs) map finite-dimensional vectors to vectors and cannot generalize to arbitrary loading histories. We employ **Deep Operator Networks (DeepONet)**, which approximate nonlinear operators mapping input functions (strain history) to output functions (stress response) [2].

1.2 Problem Statement

We model a viscoelastic material governed by the Maxwell constitutive law. The objective is to learn the operator $\mathcal{G}$ that maps an arbitrary strain history $\varepsilon(\tau)$ (for $\tau \in [0, t]$) to the current stress $\sigma(t)$:

$$\sigma(t) = \mathcal{G}[\varepsilon(\cdot)](t) \quad (1)$$

The surrogate must be:

1. **History-Dependent:** Capable of capturing memory effects (relaxation).
2. **Resolution-Independent:** Capable of predicting stress at any continuous time t .
3. **Differentiable:** Providing the tangent stiffness $\partial\sigma/\partial\varepsilon$ required for the global FEM solver's convergence.

1.3 Theory: The Neural UMAT Architecture

1.3.1 Deep Operator Network (DeepONet)

The DeepONet architecture is based on the universal approximation theorem for operators. It consists of two parallel subnetworks:

- **Branch Net:** Encodes the input function (strain history) discretized at m sensors $[\varepsilon(t_1), \dots, \varepsilon(t_m)]$ into a latent feature vector $\mathbf{b} \in \mathbb{R}^p$.
- **Trunk Net:** Encodes the coordinate (current time t) into a basis vector $\mathbf{t} \in \mathbb{R}^p$.

The operator output is the dot product of these two vectors:

$$\mathcal{G}[\varepsilon](t) \approx \sum_{k=1}^p b_k(\varepsilon) \cdot t_k(t) + b_0 \quad (2)$$

This separation of variables allows the network to learn the "memory" of the material (via the Branch Net) independently of the temporal query (via the Trunk Net).

1.3.2 Universal Training Strategy

A critical challenge identified during development was the *Out-of-Distribution (OOD) Shift*. A network trained only on random sine waves failed (104% error) when exposed to the monotonic "ramp" loading characteristic of FEM simulations. To resolve this, we implemented a "**Universal**" **Data Generator** comprising:

- **Harmonic Bursts:** To learn frequency response.
- **Random Walks:** To learn arbitrary path dependence.
- **Linear Ramps:** To learn the specific start-up transients of quasi-static loading.

1.4 Results and Analysis

1.4.1 Accuracy and Generalization

The DeepONet trained on the universal dataset achieved a **Relative Error of 3.89%** against the analytical Maxwell solver (refer to Figure 3 in project files).

- **Phase Alignment:** The network correctly captures the phase shift between stress and strain, a hallmark of viscoelasticity.
- **Ramp Handling:** Unlike initial iterations, the final model accurately predicts the stress evolution during the initial $t = 0 \rightarrow 0.1s$ loading phase, validating the robust training strategy.

1.4.2 Computational Speedup

We integrated the trained DeepONet as a Neural UMAT (User Material) subroutine. Benchmarking on a 1D bar problem with 500 elements revealed:

- **Classical Solver:** 2.10 seconds (iterative ODE integration).
- **Neural Surrogate:** 0.09 seconds (batched matrix multiplication).
- **Speedup Factor: 22.2x.**

This speedup confirms that DeepONets can effectively break the computational bottleneck of concurrent multiscale modeling [1].

References

- [1] Frédéric Feyel and Jean-Louis Chaboche. Fe2 multiscale approach for modelling the elastoviscoplastic behaviour of long fibre sic/ti composite materials. *Computer methods in applied mechanics and engineering*, 183(3-4):309–330, 2000.
- [2] Lu Lu, Pengzhan Jin, Guofei Pang, Z Zhang, and George E Karniadakis. Learning nonlinear operators via deepnet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.